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Tiny nuclear reactors are about to revolutionize American energy. Plus: the science of making nonalcoholic beer; how New Zealand is the future of wireless electricity in 2021; and more!

FEATURES

Not Much Bigger Than a Schoolbus, These Tiny Reactors Can Save Nuclear Energy

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POPULAR MECHANICS

These Tiny Reactors Can Save Nuclear Energy

BY CAROLINE DELBERT

NuScale's reactor design is 1 percent the size of a traditional plant, but packs 10 percent of the power.

NUCLEAR ENERGY'S REPUTATION IS ON the edge of meltdown. Often touted as a more flexible, powerful alternative to renewables, nuclear's advantages haven't outrun its dogged volatility. We associate nuclear reactors more often with disaster than innovation, but as the United States takes more coal and gas plants offline, engineers are hoping fresh reactor concepts could redeem nuclear's stature in American energy. Bigger is no longer better. The future, experts say, looks like "multispeed" nuclear energy—a combination of traditional large plants and smaller, safer megawatt reactors.

"Until now, customers only had one choice for nuclear, and that was a gigawatt-sized power plant," says Rita Baranwal, the assistant secretary for Nuclear Energy at the Department of Energy. "Now, we're talking about reactors at the megawatt scale that can flexibly meet a customer's energy needs as demand grows."

Baranwal says megawatt reactors (one megawatt powers about 650 homes) will be cheaper to build and operate, and could be sited anywhere in the world. A more context-sensitive, localized nuclear power industry—in which small towns, remote facilities, and big cities find nuclear solutions tailored to their needs—could replace a large swath of fossil fuel power stations, filling in the inert, resource-sucking downtimes left by renewables.

Each USNC reactor has a graphite core that houses the fuel.

THE TENETS OF NUCLEAR REACTOR design can be traced back to Enrico Fermi's original 1942 reactor at the Argonne National Laboratory near Chicago. Fermi and the other engineers at Argonne proposed and designed many kinds of reactors, like commercial boiling-water reactors and experimental research reactors, says Katy Huff, nuclear engineer and researcher at the University of Illinois at Urbana-Champaign's nuclear, plasma, and radiological engineering (NPRE) program. Argonne spearheaded a creative boom in nuclear energy. "All these different reactors were conceived at the same time," Huff says. "At the beginning, we were trying all of them."

Not every reactor survived. Just as HD DVDs fell to Blu-ray, so did experiments like the sodium reactor at the Santa Susana Field Laboratory and the gas-cooled Peach Bottom plant in Pennsylvania fall to the more commonplace light-water reactor. Even nuclear-reactor designs owe some success to chance, convenience, and market whim.

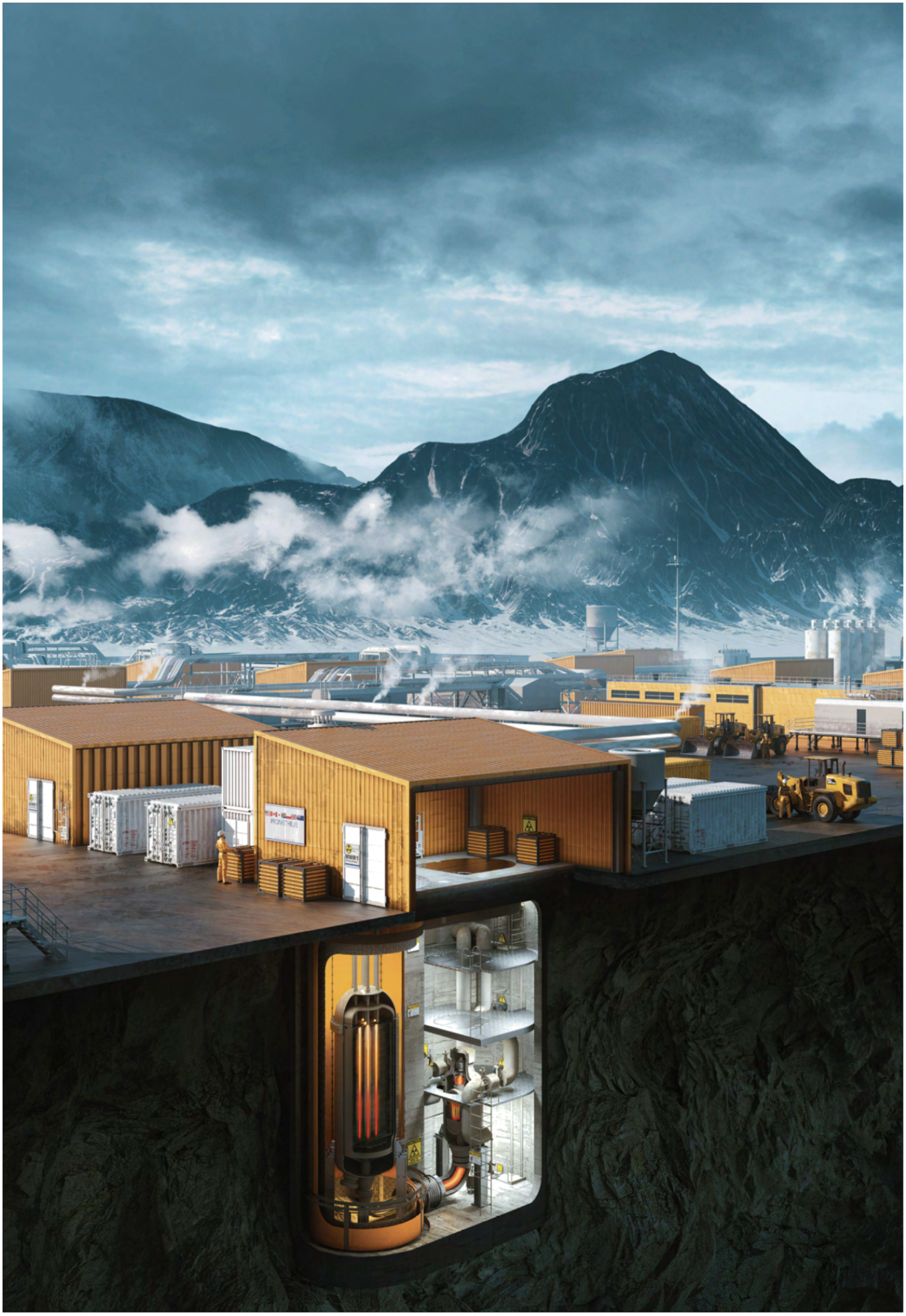
The ideal reactor by today's standards emphasizes safety, cost-effectiveness, and adaptability. Ever since engineer Richard Eckert of the New Jersey Public Service Electric & Gas Co. conceived of a floating nuclear plant in 1969 —one that could be built off-site, towed to its final destination, and operated over water—there's been a vein of thinking in nuclear that says smaller reactors are a suitable form of power for the future. That's where tiny nuclear hopes to prove itself a competitive alternative to the massive nuclear power plants we know.

"TRYING TO LOWER THE PROBABILITIES [OF ADVERSE EVENTS] IS A LOSING GAME."

Generally, tiny reactors have a few advantages over power plants. First, they are more space-efficient. A 2019 small modular reactor design from Oregon startup NuScale is about 1 percent the size of a traditional power plant's containment chamber, though it delivers 10 percent of a plant's power output. And while traditional nuclear plants need a 10-mile safety buffer in every direction in case of a meltdown, tiny reactors can operate in close quarters with much less risk. That's the second major edge: safety. Some small reactor designs incorporate fully passive safety and risk-management systems that rely on the steady, immutable laws of physics—like gravity or buoyancy—to perform safety functions rather than the actions of people or mechanical equipment. Finally, there's scalability. Many small reactors are designed with replicability in mind. If a community, factory, or city needs an extra boost of nuclear power, they can just order another reactor.

NUSCALE IS THE MOST CONSERVATIVE of the small-nuclear programs trying to reach the market. Their reactor design is a traditional light-water reactor (see image above), but smaller, simpler, and—according to NuScale—more scalable. The design eliminates coolant pumps, external steam generator vessels, and other equipment found in existing plants, so NuScale claims it's cheaper to manufacture, lower-risk for operators, and easier to maintain.

The light-water design gives NuScale an edge in power among its tiny-reactor peers. While other companies are seeking approval for reactors that produce just a few megawatts of energy, NuScale's design reaches 60 megawatts. For comparison, the smallest nuclear plant in the U.S. produces 600 megawatts, yet it's 100 times the size of NuScale's power module.



USNC's reactor produces 15MW of thermal power and 5MW of electric power.

NuScale's reactor will seek to complement, not replace, a grid already dependent on renewables. During the day, it can operate at 20 percent energy production to let the renewables do the work, before cranking up to 100 percent at night, while the rest of the grid is on solar downtime.

KATY HUFF AND UIUC ARE seeking approval to build their own tiny reactor based on designs from the Ultra Safe Nuclear Corporation (USNC), a company taking a more extreme approach than NuScale in an effort to push the field toward new standards of safety and more possible use cases. USNC's proposed design is rooted in low energy density and low decay heat generation after shutdown, which means less risk of a meltdown. The energy startup is mindful that volatility has doomed other nuclear efforts in the past. "Normal reactors are in the 20 to 40 watts per cubic centimeter power density," USNC founder Lorenzo Venneri says. "We're in the 1 to 3 [watts] per cubic centimeter power density."

"Fuel for nuclear reactors was developed for nuclear submarines where demands are entirely different from demands of a power plant," Venneri says. "A submarine is like a high-velocity sports car: It needs to go up and down in power very quickly. That's exactly the opposite of what a nuclear power plant should be for producing power."

"THE INDUSTRY TAKES THIS INCREMENTAL APPROACH. WE HAVE TO DO THINGS MORE TRANSFORMATIVELY."

USNC's reactor concept uses Fully Ceramic Micro-Encapsulated fuel (FCM), in which a ceramic-carbon and silicon-carbide composite coats granules of uranium oxide. The ceramic protects the grains of fuel, but still conducts heat. When this fuel is used in a low power-density environment, it creates a reactor that, according to Venneri, can't melt down. Inside the reactor, a feedback mechanism stops the reaction when it exceeds operating temperature, so nothing can become hot enough to melt. This contrasts an established pattern in nuclear plant design, Venneri says, to "push the envelope and then build another envelope around it."

Engineers like Venneri calculate risk as the product of probabilities and consequences. "We found out that trying to lower the probabilities is a losing game," he explains. "We want to try to keep the probabilities of adverse events low, but at the same time, make sure the consequences are zero."

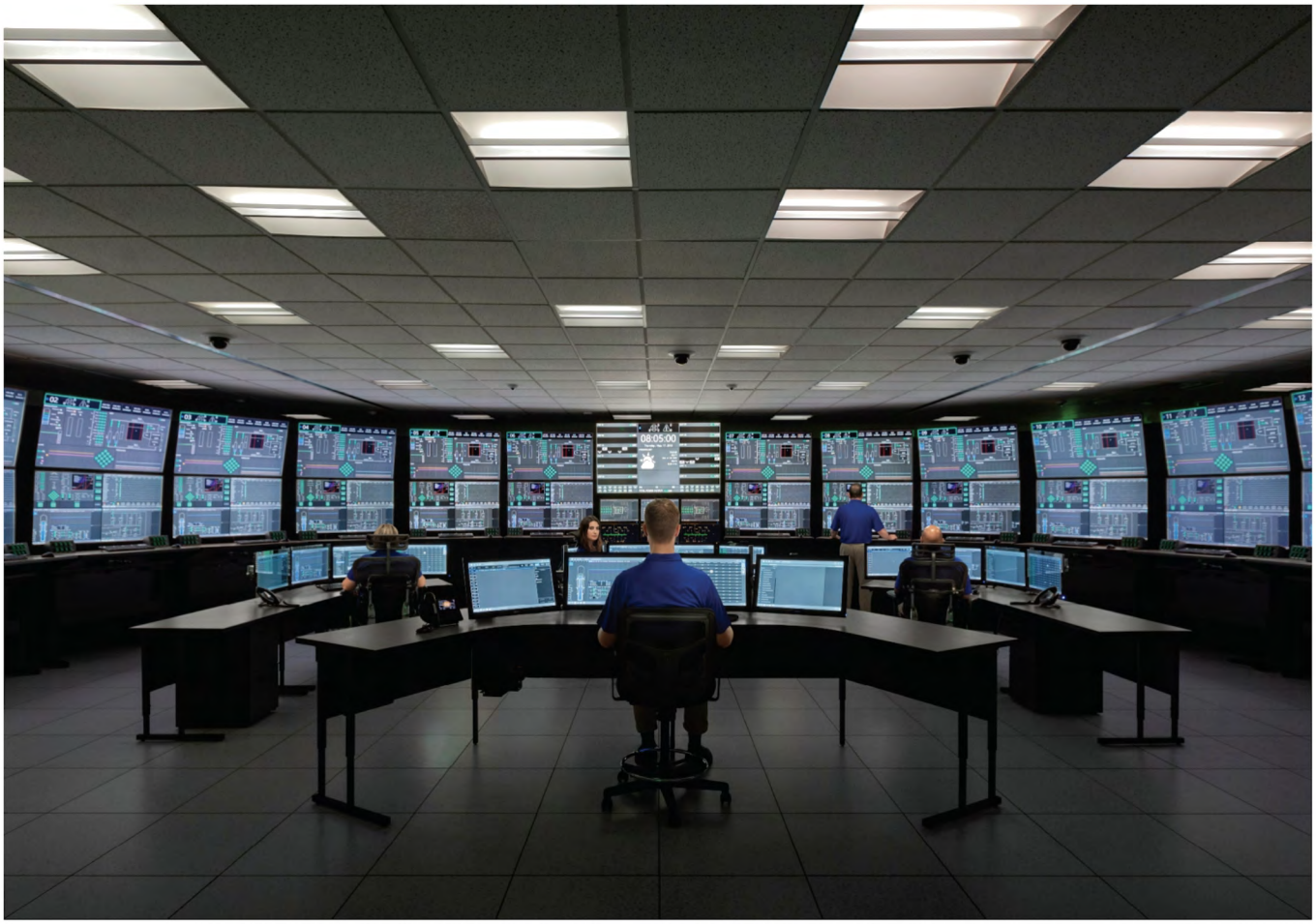
THE CALIFORNIA-BASED STARTUP OKLO IS taking risk prevention even further, to the point where they're eschewing America's legacy of water-cooled reactors. Their advanced fission microreactor can use sodium as a coolant (among other methods), so it doesn't require water. The 1.5-megawatt plant, Oklo says, functions much like a battery in producing electrical power. It's self-contained, and it can self-sustain for 20 years without refueling.

"The industry typically takes this incremental approach to things," says Jacob DeWitte, Oklo cofounder and CEO. "But in our mind we have to do things more transformatively, frankly for the planet's sake."

The Oklo reactor uses high-assay, low-enriched uranium fuel (HALEU) to achieve greater efficiency and power-per-volume given its size. HALEU is enriched to between 5 and 20 percent of the isotope U-235—the isotope that, when split, produces heat in a nuclear reaction—compared to the 3 percent low-enriched uranium in a typical power plant. This potential groundbreaker has more legwork ahead for approval than its water-cooled cousins, but Oklo is banking on its undemanding spatial and financial design. Its demonstration site at the Idaho National Laboratory spans just a quarter-acre. "We're still fission, but other than that, we use different fuel, different cooling, different technology," says Oklo cofounder and COO Caroline Cochran.

IF THE CURRENT NUCLEAR ENERGY infrastructure were a circulatory system, only the major veins and arteries would be pumping—those are the giant power plants. But tiny reactors can be like capillaries, extending power to the extremities (small towns, remote industry encampments, tiny islands, and specific city blocks) of the nuclear-power body.

"If you look at current big reactors and where they're being installed, they're going into countries that have a need to decarbonize," Westinghouse Electric Company CTO Ken Canavan says. Countries like China and Poland are "replacing multiple medium or large coal plants with single nuclear," he says, and these countries have contexts where large nuclear plants are appropriate. "If you look at other countries that have smaller grids, they don't have the capacity to put on a big nuclear plant." That's where reactors like the one at NuScale—just 65 feet tall and 9 feet in diameter—can be of service.



A plant using NuScale's tech is expected to be fully operational by 2030.

Canavan does hedge, however, to say traditional reactor concepts could make a comeback after tiny nuclear hits retail. He says tiny nuclear might disrupt nuclear energy beyond what we can project today, and in a future where more things will need electrical power (not the least of which will be your car) than ever, a "multispeed" or "multidimension" nuclear energy market is likely to emerge. Here, anyone wishing to replace a remote diesel microgrid can find a solution in nuclear, or a combination of nuclear and renewables.

For now, Oklo, USNC, and NuScale focus on small markets as their entry point because in the United States, large plants are either aging out or are too stigmatized to be replaced. Finding common but untapped use cases for tiny nuclear, like rural towns, will be key to reinvigorating interest in nuclear power technology, and finding ways to integrate nuclear alongside or into renewable-focused grids will be important to acclimating the world to nuclear. A 2020 University of Sussex study found that the development of large nuclear plants stanched wide portions of global renewable-energy development between 1990 and 2014, but another 2020 report on the near future of energy showed four separate scenarios that demonstrated runaway growth in renewables alongside modest growth of nuclear energy. We see these things as competitors; they could coexist.

By 2040, if these small reactor projects are successful, we could see a selection of plant sizes, technologies, and types of locations enter the American energy landscape. "If you look at the way everything is going, it's about personalization," Canavan says. "What's that worth to us? The capacity [small reactors] offer, and the capability they provide, is just irreplaceable."